

## Homework 1

*Due: October 2, 2026 (11:59 PM ET)***Instructions**

1. You can collaborate with one other classmate, but the final write-up/code should be your own. Acknowledge the person you've worked with.
2. You are allowed to refer to the lecture slides and any links posted on the course website for reference. **DO NOT USE** any AI tools or other online resources.
3. All submissions must be made via Gradescope (on Canvas).

**1 Breaking the Vigenère Cipher**

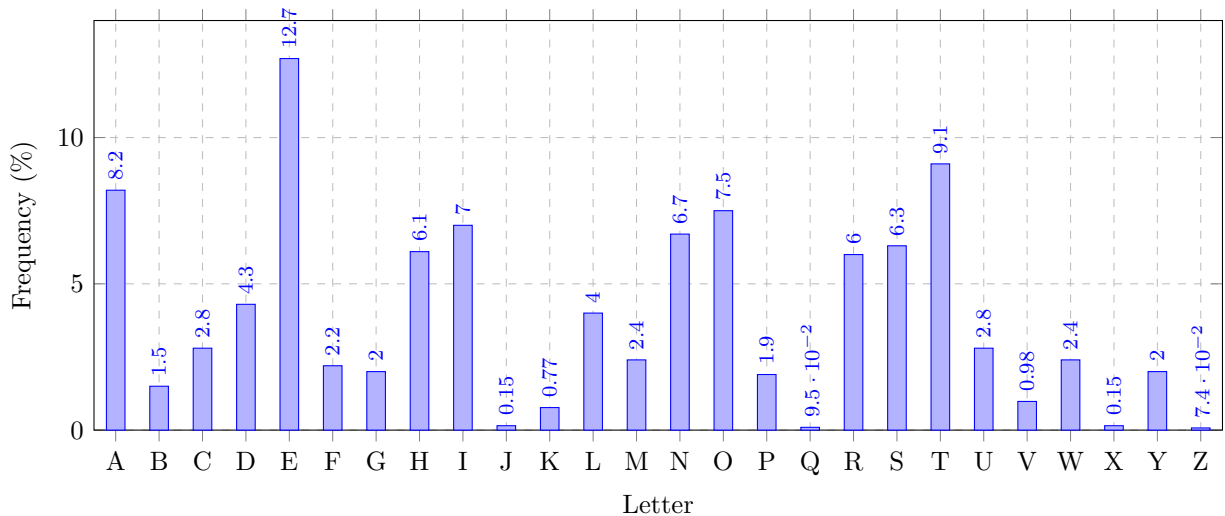
**(15 points)** The following ciphertext was encrypted using a standard **Vigenère cipher**. The plaintext consists only of English letters. Before encryption, all spaces and punctuation were removed, and all letters were converted to uppercase. The spaces in the ciphertext below are included only for readability and are not part of the ciphertext. You are **NOT** given the key or its length.

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ERNWM KVZNN EGZCS EELLR IPBRX RLFPD SHVKU ZUFTR UFZFL XEIFE OCKSK RNTZB PZGLR LWMGY
ZMNM TNYDC HPZGT CNEGF ZXUAB GRGTO AHRVB QTUXI REWIR LXNRW LXGSN APXNF TCRDI FMLVP
LEFTV TVAHR KRRZP DNYXV EUUYB YJPLE FTVNU ZIFLE ZQEOU TZVHD EQTWZ ZALRE IKGPR FAMWG
EOCKS KRNTZ BPZGL RLFIJ FLGRL MEGSI FVMGU PRRTG YYPTG XVZFC ECEET ROBLT RFGSE EEIKG
PRNYM ORONH FFVEZ FCHWZ GTOAL ENNJI AMLVN WPUTF VGLSP HQDHY IPTXZ BYBRV EDRXO EXMDC
ZRGTR KZZRR LSGUT SGBGR GPDFN FJGTT HMMFA NICAI IFHEE XHVIP LBIIU VYAZH RFNWP UTFVG
TCFNF JGTHH MMFAN ICAII RLCUI PRVYT RQXCR ETRKM JEPY TGVMQ YNYMO ROCVI LVEEE KMPVG
EEEML VFPCV ILVED WRKIN VOEYR YJROI AWMGY ZMNM TPZRE XWGBY DRGGV NYDZB PZGLR LVSDZ
FNVVE KVZNO NXXUP YJXVV RGEAM YRYWY FASNA EOOXZ LYYEE TFCRE OSKIH HPNPR EENWY FBWSR
NAHLI JBKEY XXXRC SBVGL EXUPA QFEPO SMIEG SAAHX YRCVS GIETW IFAXF ZLKRY VVDFE AVCRA
LLLLM JZZRR WMWST CHEXG BWYNE TYNME GBGTV AHRKW NRCEV GXIBO UPXHK UPVVZ IERCE PBTYR
CDROI CBAEQ YVFZT DRTWU NEIAZ XFGSE FBBKR PNGAG VAEUE RYJRD AEXTV NEIAZ OVLHO EWFN
APYRH ZSQEE XRKPL EFTVJ UTFGL XFQTF SXVVA ELRMX VEDOS MLVZP SFTKV VEWNL YJROF BKHZC
WOZTX ZPLNQ FMCVE AERGF ZXUAB GRGTO ATRUJ LSBGG VPZNF BHVEP DRQGV CEIBG ECYJD VYJZP
FLGMS SEPAX ASNRG EEMIT UYIDN IJOLS RWSEE PPRTX VQAAG MIIAD AAWPV GEEY VVDFE AVMV
PVRGX LNWLL FEURT TCHWJ VMLRM SIRNO IXVSB EHGAI BRJLR GKKUL NQMLV XPYOR XYRYI AXXVR
YTUTR URLRY RXNRY TVXXY PPNGN VZRDG BOIIA XEAMW RAOMV EMKNC IRLME PCENL METWY EXPZR
OOAFI TULNV VECNY DREIT GCOZX GYNYI PTPTV AHRKW PFEEZ LHLET NTPSI YOWNK MZEZT BKQRP
SIAWX JHNHN LXRYR EEFEE RYITF EDNNH VGINR CEHLI UGZPE HXVPE LNKKV IZLHF IJBQM VEMKN
CYPHQ DHYIP TXZBY EABKD NNOAM MEHLI YRGYN YGRWX YRDUO LXZGF TVHRR CALVX HKBPA PAPVG
EEEFE BVYGV MQLPS MBKIJ BAHVL XZPLT RXYNY YCYTW JVNAY LYSFE IGNXZ BYCVI LVEDN ROIIG
SEYXW JPCYC MEENW YFMWN RCERO IEGFA YECRO WEGHI OCWOV MQRGS EZTXZ PLLFM VLPEU EXSGR
CAGHV DVDTN DIJNY DCKIU VNTNU PVZPS FTKVS ZRZTX JGZBE XEBFT GABJZ PLNGT QFHYT FHJVA
NRLIX VQERN YJZPE RNWMK VZNNE GZCSE ELTCN JEQTR ZZAOE MEEGC OYXME GSEUB WKBCY BYWVP
FRRVS DZFNV VEKVZ NONXD BDTBY XYRXW RKIUR DITGI UOPFB KIDBO EEGGF ZAUGX VJNYD OXJFE
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|       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| PCERT | KBRN  | ILPJL | SFMYU | VPDHL | METQO | EFECZ | LTUXQ | RGTGN | EWVPF | RVMCU | RQIAB | XZBYS | GAIZE |
| HENDR | VFDEF | AICCP | DZHXZ | ILTRM | LVQPV | RESGZ | PNGHJ | DBOEE | GGILA | TBZVR | CSY   |       |       |

Recall the algorithm discussed in class for breaking Vigenère ciphers. **Using this algorithm, recover both the encryption key and the underlying plaintext. If you choose to solve the problem manually, explain your cryptanalysis in detail, including how you determine the key length and recover the individual key characters. Alternatively, if you write a program to perform the attack, include either your code or clear pseudocode describing your approach.**

For reference, the graph below shows the approximate frequency with which each letter occurs in typical English text.



## 2 Perfect Secrecy

- (5 points) Alice and Bob share a secret key  $K = 0xFFEEDD$  (written in hexadecimal). Alice wishes to privately send the message  $M = 0x012345$  to Bob, and she can only send ciphertexts written in hexadecimal.

**Explain how Alice can adapt the XOR-based one-time pad encryption scheme discussed in class (defined over binary strings) to encrypt the message  $M$  using the key  $K$  and send the resulting ciphertext to Bob in hexadecimal form. Also specify the ciphertext that Alice sends.**

- (10 points) Determine whether the following scheme is perfectly secure or not. If it is a perfectly secure encryption, prove that it satisfies both correctness and perfect security. If not, explain which of these properties is violated.

The message space is  $\mathcal{M} = \{0, 1, 2, 3, 4\}$ .

- $\text{KeyGen} :=$  Selects a key  $k$  uniformly at random from the key space  $\mathcal{K} = \{1, 3, 5, 7, 9\}$ .
- $\text{Enc}(k, m) := (m + k) \bmod 5$ .
- $\text{Dec}(k, c) := (c - k) \bmod 5$ .

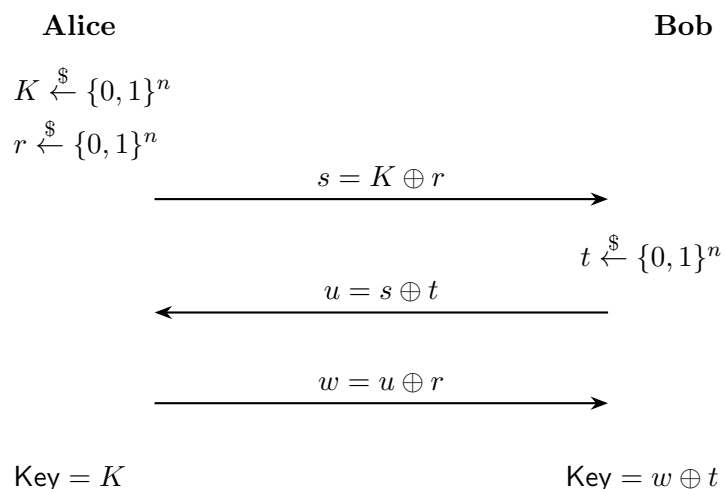
### 3 Modes of Operation

Let  $f_k$  be a *pseudorandom permutation (PRP)*, where  $f_k : \{0, 1\}^n \rightarrow \{0, 1\}^n$  and  $k \in \{0, 1\}^n$ . We interpret  $n$ -bit strings as integers in  $\mathbb{Z}_{2^n}$ , and all additions below are performed modulo  $2^n$ . Consider the following mode of operation (a variant of counter (CTR) mode discussed in class) for encrypting arbitrarily long messages  $m \in \{0, 1\}^{nt}$ , where  $t$  is arbitrary:

- **KeyGen**  $\rightarrow k$ : Sample a key  $k \xleftarrow{\$} \{0, 1\}^n$ .
  - **Enc**( $k, m$ )  $\rightarrow c$ : Parse  $m = m_1 \| \dots \| m_t$ , where each  $m_i \in \{0, 1\}^n$ . Sample a random  $\text{ctr} \xleftarrow{\$} \{0, 1\}^n$ . For each  $i \in [t]$ , compute  $c_i = f_k(\text{ctr} + i + m_i)$ . Output  $c = (\text{ctr}, c_1, \dots, c_t)$ .
1. **(5 points)** Describe the corresponding decryption algorithm.
  2. **(10 points)** **Is this scheme computationally secure?** If you believe it is secure and the key can be safely re-used to encrypt many different messages, provide an intuitive explanation (a formal proof is not required). If you believe it is not secure or the key cannot be re-used more than once, describe a concrete attack that violates computational security.

### 4 Key-Exchange

**(10 points)** Consider the following protocol between Alice and Bob:



**Is this a secure key-exchange protocol?** If so, explain why. If not, describe an attack that an eavesdropper can use to learn the final key  $K$ .

### 5 Secure Messaging

1. **(5 points)** Imagine a new law requiring communication service providers (e.g., Signal or Meta) to *escrow* all message-decryption keys by encrypting them under a law-enforcement agency's public key. This would allow the agency to decrypt any message when authorized, for

example, after obtaining a warrant or court order. **Which important security properties of end-to-end encrypted messaging would be weakened or restricted by such a requirement? Explain your answer.**

2. (5 points) **Which security properties of Signal fundamentally rely on ratcheting? Why does Signal use two different ratcheting mechanisms – the symmetric-key ratchet and the Diffie–Hellman ratchet – rather than a single ratchet?**